Received: 25 March 2024 | Accepted: 6 June 2024

DOI: 10.1111/ijlh.14330

REVIEW

ISLH International Journal of Laboratory Hematology WILEY

Hematological cytomorphology: Where we are

G. Zini[1,2]

[1]Catholic University of Sacred Heart Rome, Milan, Italy

[2]Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Hematology Institute, Rome, Italy

Correspondence
G. Zini, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Largo Agostino Gemelli 8, 00168 Rome, Italy.
Email: gina.zini@unicatt.it

[Correction added on September 22, 2025, after first online publication: The copyright line was changed.]

Abstract

The manuscript discusses the historical evolution of observing blood cell morphology under an optical microscope, from the earliest microscopes in the 17th century to the modern digital era, highlighting key advancements and contributions in the field. Blood has historically held symbolic importance in various cultures, with early medical observations dating back to Hippocrates and Galeno. The discovery of cells and subsequent advancements in microscopy by scientists like Hooke and van Leeuwenhoek paved the way for understanding blood cell morphology. Influential figures such as Hewson, Donné, and Ehrlich followed. Diagnostic cytology evolved from manual cell counting to the development of automated hematological systems. Automated complete blood counting came to support microscopic examination in diagnosing hematological disorders. Morphology is crucial in predicting disease outcomes and guiding treatment decisions, particularly hematological neoplasms. The introduction of flow cytometry and its integration with traditional morphological analysis and the new cytogenetic and molecular techniques revolutionized the classification and prognostication of hematologic disorders. Digital microscopy has emerged as a powerful tool in recent years, offering rapid acquisition and sharing of blood cell images. Integrating Artificial Intelligence with digital microscopy has further enhanced morphological analysis, improving diagnostic efficiency. We also discuss the prospects of AI in preclassifying blood cells in bone marrow aspirate samples, potentially revolutionizing diagnostic pathways for hematologic diseases. Overall, the manuscript provides a comprehensive overview of the historical development, clinical significance and technological advancements in observing blood cell morphology, underscoring its continued relevance in modern hematology practice.

KEYWORDS

general hematology, leukemia, morphology

Die as much as necessary, without exceeding.
Reborn as much as necessary from what has been saved.
Wisława Szymborska

1 | INTRODUCTION

In human history, technological innovation has always played a decisive role in acquiring new working methods. It is a continuous, unstoppable process in which the new, which advances knowledge, realizes that the old performs an irreplaceable function and is reinterpreted and repositioned.

This concept fully applies to observing the morphology of blood cells under an optical microscope (OM).

2 | PAST TIMES

Blood has always played a mysterious and symbolic role in different civilizations, regardless of geographical locations, and cultural systems. In Western thought, the first medical-psychological approach dates back to Hippocrates (1st century BC), who placed blood among the four humors responsible for the states of health and disease. Galeno (1st century AD) discovered that the arteries contained blood and not air, contributing to the transformation of ancient medical art into a science based on observation and rational explanation of phenomena.[1]

3 | DESCRIPTIVE CYTOMORPHOLOGY

The discovery of the cell would not have been possible without advancements in microscope science.[2]

The first microscopy instruments were produced in the Netherlands. In Middelburg around 1600, a Dutch eyeglass maker, Zacharias Janssen, likely made one of the first compound microscopes that used two lenses and magnified an object up to 20 or 30 times its standard size.

In 1665, the English botanist Robert Hooke looked at a piece of cork with an improved compound three-lense microscope, with a magnification up to 50 times. He called "cells" the tiny units of the cork, resembling pores, in analogy with the "plain rooms in monasteries where monks lived." The small magnification prevented him from evaluating the internal structures of these "cells," but the discovery of these "cells" and the proposed term have dramatically impacted science.[1]

Another Dutchman, Antonie van Leeuwenhoek, created a single-lens microscope by grinding his lenses to magnify the observed objects 300 times. In 1661, for the first time, he observed and described red blood cells and spermatozoa, named "living animalcules."

William Hewson, a British anatomist included among the pioneers of hematology, first, in the mid-18th century, studied and described the form, shape and function of red blood cells, white blood cells, the lymphatic system, thymus, and the spleen.[3]

The morphologic era of hematology started in the mid-19th century: the first binocular OM was developed, and the newly invented microphotography could obtain significantly larger photographic images of the respective objects.

Alfred Donné, a French bacteriologist and doctor considered the inventor of microphotography, published a "Course in Microscopy" in 1844. Among other things, he describes with great insight a maturation arrest of the white blood cells in one 44-year-old woman with splenomegaly, who probably died due to a blast crisis of chronic granulocytic leukemia. He pioneered the use of OM and trained many physicians in microscopy.

In 1845, Gabriel Andral, a French anatomy-pathologist, wrote the first treatise on Hematology, in which microscopic analysis is applied to study blood, both in health and disease. Thanks to the OM, microscopic anatomy and hematology were born. Leukemia was discovered and described almost simultaneously: in 1845, within a few weeks of each other, John Hughes Bennett in Edinburgh and Rudolph Virchow in Berlin independently reported "the presence of colorless or white corpuscles" in the blood of two dead patients. Later, in 1847, the word leukemia (white blood) was coined and is still used today.

In 1854, a British physician, Lionel Smith Beale, published "The Microscope and its Applications in Medicine", which carefully describes the different circulating blood corpuscles in healthy and unhealthy individuals.

In 1868, German pathologist Ernst Neumann defined a direct relationship between bone marrow (BM) and hematopoiesis in terms of both health and leukemia.

In 1876, Mosler used a wood drill to aspirate BM particles from a patient with leukemia, introducing BM puncture for an appropriate diagnosis of leukemia postmortem.

Paul Ehrlich, a German microbiologist and immunologist, received a Nobel Prize in 1908 for developing chemotherapy and invented tissue staining methods to distinguish and classify the different types of cells circulating in the peripheral blood (PB).

In 1900, the Swiss hematologist Naegeli identified myeloblast and lymphoblast as the ancestors of granulocytes and lymphocytes, respectively.

At the turn of the century, the concept of the existence of stem cells capable of self-renewal and differentiation into different cell types was acquired. Still, there is much debate about whether it is a "common stem cell," as argued in the Virchow school, or if precursors for leucocytes and lymphocytes differ, as Erlich sustained in the dualistic theory.[4,5]

4 | DIAGNOSTIC CYTOMORPHOLOGY

The development of microscopic techniques and blood cell staining methods became powerful instruments in understanding acute and chronic leukemias, a story that took up the whole of the 20th century.

In 1913, four types of leukemia were identified: chronic lymphocytic leukemia, chronic myelogenous leukemia, acute lymphocytic leukemia, and acute myelogenous leukemia.[6]

In 1953, Frits Zernike, a Dutch physicist, was awarded the Nobel Prize for his invention of the phase-contrast microscope. This microscope makes it possible to look at living cells and bacteria, including the processes of cell division. Until the first half of the 19th century, the colorimetric determination of hemoglobin measurement was manual, as counting blood cells in PB at the microscope.

Maxwell Myer Wintrobe, an Austrian-born American physician, develops an instrument called "the Wintrobe hematocrit" and, using mathematical formulas, produces the "Wintrobe RBC indices", MCV, MCH, and MCHC, providing the first careful definition of typical values for normal red blood cells and the first standardized diagnostic approach for diagnosis of anemia.[7] He invented the modern role of the hematology laboratory, placing it as the basis of clinical diagnostics. His textbook, Clinical Hematology, first published in 1942, is adopted worldwide: in 2023, 80 years later, the 15th edition was printed as testimony to Dr Wintrobe's legacy.

Until the first half of the 20th century, counting blood cells in PB is carried out using calibrated cameras. Cell morphology and cytochemistry evaluation are the only diagnostic techniques available for identifying normal and pathological blood cells. The complete blood count is an exam with a fully understood clinical value: cell quantification and morphology are related to clinical manifestations and symptoms. However, in this phase, the differentiation and counting of leukocyte populations remains the prerogative of the OM: producing a complete blood count is certainly not quick and not available everywhere.

Since the second half of the 20th century, the hematology laboratory has had a quick and relevant technological development, the PB analysis in flow cytometry. In 1956, the first hemocytometer was placed on the market. A semiautomatic counter of PB cells in fluid suspension that uses the "Coulter principle": blood cells are identified and counted based on the difference in electrical conductivity. In the 1970s, this technological revolution was completed by introducing quantitative cellular cytochemistry using peroxidase, lipase, and alcian blue activity.[8]

Automated hematological systems have followed one another over the years, increasing the complexity of analysis and the richness of clinically useful delivered data. PB analysis quickly becomes the most widespread economic test mainly for the following reasons: (i) accuracy and precision of results, (ii) clinical value for screening, case finding, diagnostic orientation, and monitoring of hematologic, and nonhematologic disorders and, (iii) low-cost test.

Complete blood count today is an essential laboratory test with a fully understood clinical value: cell quantification and morphology are related to clinical manifestations and symptoms. Berend Houven prophesied the future of automation by introducing the concept of "extended differential count" (ECD): "*…replacement of instrumental reporting of abnormal cell populations by an individual counting of such cells…*", such as immature granulocytes, blasts, and nucleated red blood cells, when present, in addition to the five different cell types normally circulating in PB.[9] New roles in the laboratory were then required for the microscopic examination of PB: to provide the reference leukocyte differential for the automatic systems and to evaluate and validate those samples that show quantitative and/or qualitative abnormalities in the instrumental report. The so-called "Hematology beyond microscopy" was born.[10] For these new tasks, consensus documents, rigorous guidelines, and reference standards are published and continuously updated by international groups such as NCCLS, ICSH, World Health Organization (WHO), and International and National Guidelines.[11–14]

During the 19th century, several descriptions of microscopic observations of the BM carried out postmortem have been reported. In 1908, an Italian physician, Giovanni Ghedini, was the first to underline the diagnostic value of evaluating the observation in vivo of PB together with BM, taken from the proximal tibia. In a pioneering way, he also recommends the importance of contextual observation of the BM in aspiration and biopsy.[15] Later, in 1927, Anirkin, a Russian physician, used a lumbar puncture needle to aspirate BM fluid from the sternum.[16]

Starting in the 20th century, the improvement of BM aspiration and biopsy techniques has made the observation of cells in the BM an unavoidable test for diagnosing leukemia. A new and stimulating field of application opened up for cytomorphology under the OM: the evaluation of BM aspirates.

In this period, attempts were made to systematically classify hematological neoplasms, starting from leukemias. In 1958, the first modern classification of leukemias recognized a common hematopoietic progenitor for both myeloid and lymphoid lineage. It defined acute leukemias as "*diseases of the hematopoietic tissues due to tumor proliferation…characterized by the constant infiltration of immature and atypical cells in the various organs and tissues and by habitual but not obligatory presence of said cells in the circulating blood.*"[17] Based on a panoptic stain and two cytochemical stains, Sudan Black B and Periodic Acid Schiff of PB cells observed under the OM, acute leukemias were differentiated into hemocytoblastic, myeloid, lymphoid, and monocytic.

In 1976, the first universally adopted classification of leukemias, the French-American-British (FAB) classification, was published[18]: the leukemia diagnosis and typing were based on identifying and counting PB and BM blasts, with the support of manual cytochemical techniques. Morphology and cytochemistry were later combined with a new diagnostic tool, flow cytometry, a laboratory technique capable of identifying blood cells based on the presence/absence of lineage-specific surface markers.

In 1999, the first report on the WHO classification of hemopoietic and lymphoid tissue tumors was published.[19] It was a consensus classification using all available information (morphology, cytochemistry, immunophenotype, genetics, and clinical features) to define clinically significant disease entities.

From FAB 1976 to the 2016 WHO classification,[20] the blast percentage remains a significant factor in the OM diagnosis, subclassification, prognosis, and disease progression.

5 | MORPHOLOGY AND DISEASE PREDICTIVITY

The development of cytogenetic e molecular techniques has dramatically impacted the complexity of classifications of hematologic neoplasms and prognostication and treatments. Thanks to these techniques, numerous hematological neoplasms on which the diagnosis is based on the presence of a specific molecular alteration have been identified over time. Identifying specific morphological alterations in several of these tumors allows the morphology to acquire a vital predictive function of the molecular lesion. In the classifications of hematological malignancies published in 2022, the International Consensus Classification[21] and WHO 5th edition,[22,23] the majority of diagnostic and prognostic subtypes are based on molecular and next generation sequencing tests.

Blood cell evaluation at OM remains the reference method for the quantitative and qualitative analysis of hematological cells on PB and BM smears. However, the perspective changes: from a diagnostic tool tout court to a screening and diagnostic predictive tool. The first step in the diagnostic pathway. Therefore, the so-called "morphological diagnosis" is not the final diagnosis: it will be confirmed or refuted based on the results of those further diagnostic tests required by international diagnostic guidelines.

The unique characteristics of the morphology under the OM, as well as immediacy and cost-effectiveness, allow for the timely and rapid identification of patients who require further appropriate diagnostic investigations. Nevertheless, the lack of standards in the preparation and staining methods of PB and BM smears, the absence of a universally shared identifying nomenclature of blood cells, and the subjectivity linked to individual skills and experience are crucial aspects that impact intra- and inter-operator variability that negatively influences and limits the diagnostic value of morphology.[24,25] Furthermore, working at the OM is physically tiring and requires time for training, impacting the availability of human resources to dedicate. Indeed, in this scenario of rapid technological progress, microscopy's value and efficacy in the diagnostic work-up of hematological disorders is periodically debated, up to the proposal for clinical hematologists "to put the OM on e-bay."[26]

The actual predictive capacity in diagnosing anemia, leukemia, and infections is confirmed in the scientific literature, which, to this day, in the postpandemic era, offers indicative screening based on qualitative and/or quantitative anomalies. Identifying and enumerating both normal and abnormal blood cells constitute the cornerstone: it is the starting valuable point in all the contexts of screening, diagnosis, and monitoring of both hematological and nonhematological diseases with PB and BM involvement.

6 | CYTOMORPHOLOGY AND TECHNOLOGICAL REVOLUTION

In recent years, morphology has found an unexpected and formidable amplification due to the development of DM applied to hematology. In fact, the digital format has allowed the rapid diffusion and easy use of blood cell images, playing the same role in transmitting knowledge of blood cell morphology played by the printing press invented in Germany in the 15th by Johannes Gutenberg. PB smear images of exceptional quality can be acquired rapidly and conveniently. Images are immediately available for incorporation into websites or digital publications, printing, and transfer to other individuals by e-mail or other applications. These images are essentially indistinguishable from conventional film images. Diffusion of this technology establishes a virtuous circle that continuously feeds the diffusion, training, and maintenance of the morphological skills essential for its validation. The computer and the digital camera represent effective and robust technical tools for facilitating and improving hematology education, research, and patient service.[27,28] The development of AI technology applied to DM has triggered a formidable, unprecedented revolution in the use of blood cell morphology.[29] Several AI systems applied to DM are a reality in our laboratories, providing excellent PB cell preclassifier systems. The more they spread, the more those limitations inherent in generating the report at the OM on the smear glass will be definitively resolved. Consultancy, remote review, easy cell image retrieval, reduction of fatigue, costs, number of reviewing at OM, easy improvement of intra- and inter-laboratory morphology education, harmonization, and peer reviewing. Cells are automatically preclassified, not based on their physical–chemical properties, as it occurs in flow cytometry counting: the whole process of optical and cerebral connections that allow cells to be recognized at OM now takes place on the screen. A true rebirth of morphology is understood as identifying blood cells based on size, shape, structure, and shades of color: a slow and continuous unstoppable process of diffusion of morphological competence.[30–32] Representative examples of the gamut of opportunities provided by machine learning for cell classification and diagnostic orientation include, among others, the identification of dysplastic features, blast cell characterization and enumeration, red blood cell morphological abnormalities, and malaria detection.[33,34]

In technological development, some drawbacks cannot be ruled out even when AI is applied to morphological diagnostics.[35] At the most basic level, it is possible that in daily operations, even the most advanced methodologies produce conflicting conclusions or biases on the interpretation of some rare and challenging cells or, on the contrary, discriminatory conclusions on elements very similar to a regular counterpart.[32] Selective scanning of areas of the smear may limit the identification of elements hidden in the margins and tails. Moreover, elaborate AI devices could limit the opportunities to morphologically identify unexpected details or connections. Especially models based on deep learning, which effectively function as black boxes for operators, leave no room for interpretability and understanding of decision-making processes in cytomorphology. Some of these limitations have already been highlighted in the Recommendations of the International Council for Standardization in Hematology.[30]

The new challenge concerns the preclassification of cells in BM aspirate samples.[36] Currently, a stimulating scientific production exists that evaluates the performance of some AI-based preclassifiers already available to BM.[37] This means that this goal is also now very close. From micrography through BM microphotography, we have now arrived at digitized reproduction to achieve the same goal: identify and count blood cells to allow our patients a rapid and effective clinical diagnostic pathway.

CONFLICT OF INTEREST STATEMENT

The authors have no competing interests.

PATIENT CONSENT STATEMENT

The study used archived material. No material from other sources was used. The study was not a clinical trial (no registration required).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the general literature databases (i.e., Pubmed).

ORCID

G. Zini https://orcid.org/0000-0003-0782-294X

REFERENCES

1. Hajar R. The air of history: early medicine to Galen (part I). *Heart Views*. 2012;13:120-128.
2. Wollman AJ, Nudd R, Hedlund EG, Leake MC. From Animaculum to single molecules: 300 years of the light microscope. *Open Biol*. 2015; 5:150019.
3. Hooke R. *Micrographia: or Some physiological descriptions of minute bodies made by magnifying glasses*. J. Martyn and J Allestry; 1665.
4. Doyle D. William Hewson (1739-74): the father of haematology. *Br J Haematol*. 2006;133:375-381.
5. Coller BS. Blood at 70: its roots in the history of hematology and its birth. *Blood*. 2015;126:2548-2560.
6. Kampen KR. The discovery and early understanding of leukemia. *Leuk Res*. 2012;36:6-13.
7. Wintrobe MM. The size and hemoglobin content of the erythrocyte. Methods of determination and clinical application, 1932. *J Lab Clin Med*. 1990;115:374-387.
8. Hudson MJ, Green AE. Visual versus mechanised leucocyte differential counts: costing and evaluation of traditional and Hemalog D methods. *J Clin Pathol*. 1980;33:1114-1117.
9. Houwen B. The differential cell count. *Lab Hematol*. 2001;7:89-100.
10. Pierre RV. Peripheral blood film review. The demise of the eyecount leukocyte differential. *Clin Lab Med*. 2002;22:279-297.
11. Lee SH, Erber WN, Porwit A, Tomonaga M, Peterson LC. International Council for Standardization in Hematology. ICSH guidelines for the standardization of bone marrow specimens and reports. *Int J Lab Hematol*. 2008;30:349-364.
12. Briggs C, Culp N, Davis B, d'Onofrio G, Zini G, Machin SJ. International Council for Standardization of Haematology. ICSH guidelines for the evaluation of blood cell analysers including those used for differential leucocyte and reticulocyte counting. *Int J Lab Hematol*. 2014;36:327-613.
13. Clinical and Laboratory Standards Institute, ed. *Reference Leukocyte (WBC) Differential Count (Proportional) and Evaluation of Instrumental Methods*. 2nd ed. Clinical and Laboratory Standards Institute; CLSI Document H20-A2; 2007.
14. WHO Classification of Tumours. In: Swerdlow SH, Campo E, Harris NL, et al., eds. *WHO Classification of Tumours of Haematopoietic and Lymphoid Tissues*. 4th ed. IARC Ed; 2008.
15. Pizzi M, Binotto G, Rigoni Savioli G, Dei Tos AP, Orazi A. Of drills and bones: Giovanni Ghedini and the origin of bone marrow biopsy. *Br J Haematol*. 2022;198:943-952.
16. Parapia LA. Trepanning or trephines: a history of bone marrow biopsy. *Br J Haematol*. 2007;139:14-19.
17. Ferrata A, Storti E. *Le malattie del sangue*. 2nd ed. F. Vallardi Editore; 1958.
18. Bennett JM, Catovsky D, Daniel MT, et al. Proposals for the classification of the acute leukaemias. French-American-British (FAB) co-operative group. *Br J Haematol*. 1976;33:451-458.
19. Harris NL, Jaffe ES, Diebold J, et al. The World Health Organization classification of neoplastic diseases of the hematopoietic and lymphoid tissues. Report of the clinical advisory committee meeting, Airlie house, Virginia, November, 1997. *Ann Oncol*. 1999;10:1419-1432.
20. Swerdlow SH, Campo E, Harris NL, et al. *WHO classification of Tumours of Haematopoietic and lymphoid tissues*. 4th ed. IARC Ed; 2017.
21. Arber DA, Orazi A, Hasserjian RP, et al. International consensus classification of myeloid neoplasms and acute leukemias: integrating morphologic, clinical, and genomic data. *Blood*. 2022;140:1200-1228.
22. Khoury JD, Solary E, Abla O, et al. The 5th edition of the World Health Organization classification of Haematolymphoid Tumours: myeloid and histiocytic/dendritic neoplasms. *Leukemia*. 2022;36: 1703-1719.
23. Alaggio R, Amador C, Anagnostopoulos I, et al. The 5th edition of the World Health Organization classification of Haematolymphoid Tumours: lymphoid neoplasms. *Leukemia*. 2022;36:1720-1748.
24. Naqvi K, Jabbour E, Bueso-Ramos C, et al. Implications of discrepancy in morphologic diagnosis of myelodysplastic syndrome between referral and tertiary care centers. *Blood*. 2011;118:4690-4693.
25. Fuentes-Arderiu X, García-Panyella M, Dot-Bach D. Between examiner reproducibility in manual differential leukocyte counting. *Accredit Qual Assur*. 2007;12:643-645.
26. van't Veer M, Haferlach T. Should clinical hematologists put their microscopes on eBay? *Haematologica*. 2014;99:1533-1534.
27. Zini G, Barbagallo O, Scavone F, Béné MC. Digital morphology in hematology diagnosis and education: the experience of the European LeukemiaNet WP10. *Int J Lab Hematol*. 2022;44(Suppl. 1):37-44.
28. Burthem J, Brereton M, Ardern J, et al. The use of digital virtual slides in the quality assessment of haematological morphology: results of a pilot exercise involving UKNEQAS(H) participants. *Br J Haematol*. 2005;130:293-296.
29. Zini G. Artificial intelligence in hematology. *Hematology*. 2005;10: 393-400.
30. Kratz A, Lee SH, Zini G, Riedl JA, Hur M, Machin S. Digital morphology analysers in hematology: ICSH review and recommendations. *Int J Lab Hematol*. 2019;41:437-447.
31. Park SH, Park CJ, Choi MO, et al. Automated digital cell morphology identification system (CellaVision DM96) is very useful for leukocyte differentials in specimens with qualitative or quantitative abnormalities. *Int J Lab Hematol*. 2013;35:517-527.
32. Zini G, Mancini F, Rossi E, Landucci S, d'Onofrio G. Artificial intelligence and the blood film: performance of the MC-80 digital morphology analyzer in samples with neoplastic and reactive cell types. *Int J Lab Hematol*. 2023;45:881-889.
33. Obstfeld AE. Hematology and machine learning. *J Appl Lab Med*. 2023;8:129-144.
34. Zhang Z, Yang S, Wang X. Schistocyte detection in artificial intelligence age. *Int J Lab Hematol*. 2024;46:427-433.
35. Walter W, Haferlach C, Nadarajah N, et al. How artificial intelligence might disrupt diagnostics in hematology in the near future. *Oncogene*. 2021;40:4271-4280.
36. Zini G, Chiusolo P, Rossi E, et al. Digital morphology compared to the optical microscope: a validation study on reporting bone marrow

aspirates. *Int J Lab Hematol*. 2024;46:474-480. doi:10.1111/ijlh.14238

37. Wang M, Dong C, Gao Y, Li J, Han M, Wang L. A deep learning model for the automatic recognition of aplastic anemia, myelodysplastic syndromes, and acute myeloid leukemia based on bone marrow smear. *Front Oncol*. 2022;12:844978.

How to cite this article: Zini G. Hematological cytomorphology: Where we are. *Int J Lab Hematol*. 2024;46(5): 789-794. doi:10.1111/ijlh.14330